Assignment no. 5

Natural Language Processing
NLP4IL (AIS.ba VZ)
WS 2026

Exercise 1 Clustering of programming languages

Last assignment, you were tasked to predict the programming language based on code snippets. To get insights on what languages are similar, you will cluster the text data and visualize it accordingly.

Tasks:

1. Load the embeddings (and if needed, the text dataset).
2. Research when an if dimensionality reduction before clustering for presentation in exercise. Preprocess your data according to your findings.
3. Cluster your data with KMEANS and DBSCAN.
4. Visualize and compare and interpret the results for the presentation in the exercise.

Deliverables: A notebook with your research, clustering, visualizations, analysis and interpretation.

Exercise 2 LDA on news content You perform topic modeling on the 20 Newsgroups dataset using Latent Dirichlet Allocation (LDA). The goal is to discover latent topics in news articles and analyze how well they align with known newsgroup categories.

Steps:

1. Load the 20 Newsgroups dataset (full or selected categories)
2. Preprocess the text accordingly:
3. Train an LDA models with a varying numbers of topics.
4. Inspect and interpret topics using top-N words per topic
5. Assign dominant topics to documents
6. Visualize topic distributions using:
    - t-SNE, PCA or UMAP on document-topic vectors
    - pyLDAvis visualization

Deliverables:

- Notebook with data loading, preprocessing, and LDA training
- Discussion for the presentation:
    - How interpretable are the topics?
    - How does the number of topics affect results?
    - What are the limitations of LDA on real-world news data?

Exercise 3 BERTopic on Song lyrics

In this exercise, you will cluster Song lyrics with BERTopic. The music lyrics are usually very specific to a particular topic (think Christmas, Lovesongs, ...), is this also reflected in the resulting clusters?

Steps:

1. Load the lyrics and process them with `BERTopic`.
2. Tune the hyperparamters (UMAP and HDBScan) to get meaningful clusters.
3. Validate the clusters by taking a look at some samples of the clusters.
4. Prepare the build-in visualizations of BERTopic for presentation in the exercise.

Deliverables:

- Notebook containing the BERTopic modeling and vis.
- Presentation of the hyper parameter tuning.

When using Large Language Models (eg. ChatGPT, Copilot, ...) to produce output that is handed in, also state the prompt and model used in the submission. Submissions that are not marked and are noticeably AI generated will be graded with zero points!

Submission: electronically via Moodle; Deadline: Tue, Apr 20, 2026, 23:59.